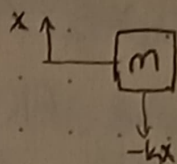


KON 313E - Feedback Control Systems - L102

Ayşe Tazari Farmani

$$m\ddot{x} = -kx \rightarrow m\ddot{x} + kx = 0$$



General solution:  $x(t) = A \sin(\omega t + \phi)$

## Fourier Transforms

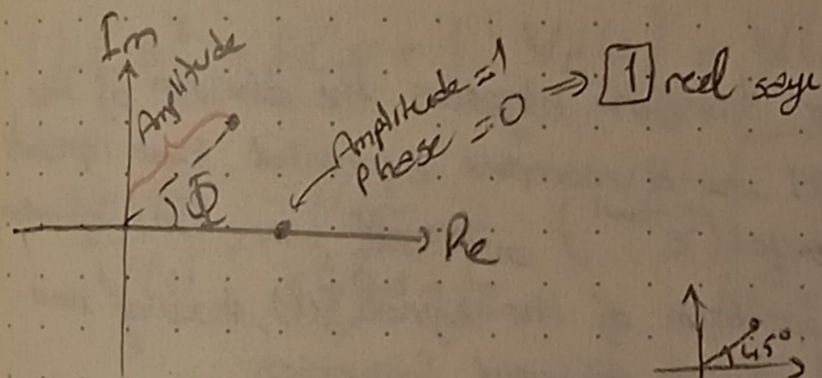
$$f(t) = \int_{-\infty}^{\infty} A(\nu) \cos(2\pi \nu t) d\nu$$

$A(\nu)$ : Amplitude at each frequency  $\nu$ .

$$f(t) = \int_{-\infty}^{\infty} A(\nu) \cos(2\pi \nu t) d\nu$$

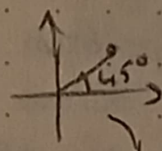
No phase info

Phase and amplitude can be described by a complex number



$$A = \sqrt{\text{Re}^2 + \text{Im}^2}$$

$$\tan \phi = \frac{\text{Im}}{\text{Re}}$$



So  $A(\nu)$  with phase info  $\Rightarrow F(\nu) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$

Euler's  $\rightarrow e^{it} = \cos t + i \sin t$

$$A(\nu) \cos 2\pi \nu t$$

with  $45^\circ$  phase shift:

$$F(\nu) e^{2\pi i \nu t}$$

$$F(\nu) e^{2\pi i \nu t} = \frac{\sqrt{2}}{2} \cos(2\pi \nu t) - \frac{\sqrt{2}}{2} \sin(2\pi \nu t)$$

$$+ \frac{\sqrt{2}}{2} \cos(2\pi \nu t) i + \frac{\sqrt{2}}{2} \sin(2\pi \nu t) i$$

Imaginary part

Positive frequency

$\frac{\sqrt{2}}{2} \cos - \frac{\sqrt{2}}{2} \sin$   
 $\frac{\sqrt{2}}{2} \cos i - \frac{\sqrt{2}}{2} \sin i$

Real part of  $F(\omega)$  is even, Im part is odd.  
 We solve  $F(\omega)e^{2\pi i \omega t}$  for negative frequency and sum with positive frequency solution.  $\int_{-\infty}^0 + \int_0^{\infty}$

$$= \frac{\sqrt{2}}{2} \cos(2\pi \omega t) - \frac{\sqrt{2}}{2} \sin(2\pi \omega t) = \cos(2\pi \omega t + \frac{\pi}{4})$$

$$f(t) = \int_{-\infty}^{\infty} F(\omega) e^{2\pi i \omega t} d\omega$$

Inverse Fourier Transform

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i \omega t} dt$$

Forward Fourier Transform

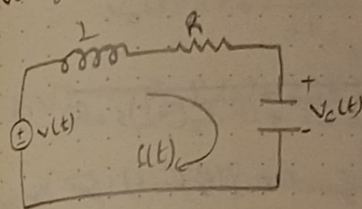
The forward Fourier transform integrates the division of the time-domain signal  $f(t)$  and a complex exponential that represents the sinusoidal frequencies ( $e^{i\omega t}$ ) over all time. This operation yields the frequency spectrum of the signal  $f(t)$ , showing how its energy is distributed across different frequencies. It tells us how much of each frequency is present at what price.

Laplace Transform

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

where  $s = \sigma + j\omega$  a complex variable

Problem: Find the transfer function relating the capacitor voltage  $V_C(s)$  to the input voltage  $V(s)$



KVL

$$L \frac{di(t)}{dt} + Ri(t) + \frac{1}{C} \int_0^t i(t) dt = V(t)$$

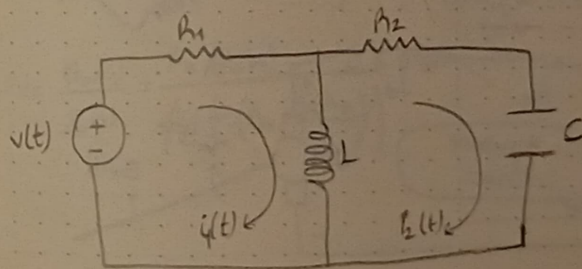
$$LC \frac{d^2 V_C(t)}{dt^2} + RC \frac{dV_C(t)}{dt} + V_C(t) = V(t) \quad \left\{ \begin{array}{l} q(t) = C V_C(t) \\ \frac{dq(t)}{dt} = i(t) \end{array} \right. \text{Take Laplace}$$

$$(LCs^2 + RCs + 1) V_C(s) = V(s)$$

$$\frac{V_C(s)}{V(s)} = \frac{1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}} \xrightarrow{V(s)} \boxed{\frac{1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}}} \rightarrow V_C(s)$$

$$[\text{sum of impedances}] I(s) = [\text{sum of applied voltages}]$$

Problem: Find the transfer function  $I_2(s) / V(s)$



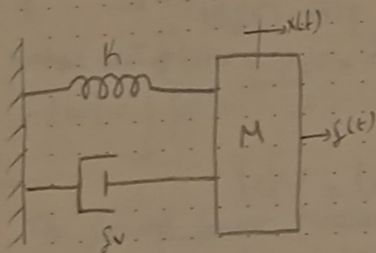


$$L_1 \dot{I}_1(s) + L_2 \dot{I}_1(s) - L_2 \dot{I}_2(s) = V(s)$$

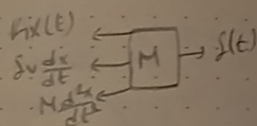
$$L_2 \dot{I}_2(s) + R_2 I_2(s) + \frac{1}{Cs} I_2(s) - L_2 \dot{I}_1(s) = 0$$

$$G(s) = \frac{I_2(s)}{V(s)} = \frac{2Cs^2}{(R_1 + R_2)L_2Cs^2 + (R_1R_2C + L_2)s + R_1}$$

**Problem:** Find the transfer function,  $X(s)/F(s)$  for the system



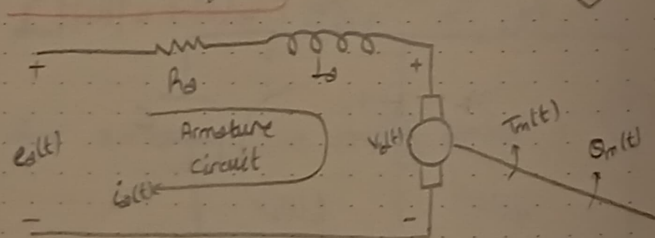
we assume the mass is travelling towards right. Spring, damper, force due to acceleration point left



$$M \frac{d^2x}{dt^2} + sv \frac{dx}{dt} + kx(t) = f(t)$$

$$Ms^2 X(s) + sv s X(s) + k X(s) = F(s)$$

$$G(s) = \frac{1}{Ms^2 + sv s + k}$$



$V_b(t)$ : back emf  $k_b$ : back emf constant

$$V_b(t) = k_b \frac{d\theta_m(t)}{dt} \rightarrow V_b(s) = k_b s \theta_m(s)$$

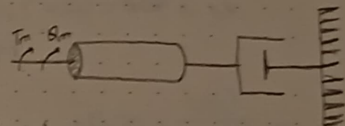
$$kVL \rightarrow R_a I_a(s) + L_a s I_a(s) + V_b(s) = E_a(s)$$

$$T_m(s) = k_t I_a(s) \quad T_m: \text{torque developed by motor}$$

$$k_t: \text{motor torque constant}$$

$$\frac{(R_a + L_a s) T_m(s)}{k_t} + k_b s \theta_m(s) = E_a(s)$$

equivalent mechanical loading of a motor



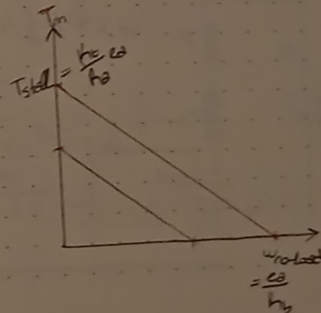
$$T_m(s) = (J_m s^2 + D_m s) \theta_m(s)$$

$$\frac{(R_a + L_a s) (J_m s^2 + D_m s) \theta_m(s)}{k_t} + k_b s \theta_m(s) = E_a(s)$$

$$L_a \ll R_a$$

$$\left( \frac{R_a}{k_t} (J_m s + D_m) + k_b \right) s \theta_m(s) = E_a(s)$$

$$H(s) = \frac{\theta_m(s)}{E_a(s)} = \frac{k_t / (R_a J_m)}{s \left( s + \frac{1}{J_m} \left( D_m + \frac{k_t k_b}{R_a} \right) \right)}$$



Begin by finding mechanical constants  $J_m$  and  $D_m$

$$J_m = J_o + J_L \left( \frac{N_1}{N_2} \right)^2 = 5 + 700 \left( \frac{1}{10} \right)^2 = 12$$

$$D_m = D_o + D_L \left( \frac{N_1}{N_2} \right)^2 = 2 + 800 \left( \frac{1}{10} \right)^2 = 10$$

Then electrical constants  $k_e / R_a$  and  $k_b$  from the curve

$$\begin{aligned} T_{stall} &= 500 \\ \omega_{no-load} &= 50 \\ e_g &= 100 \end{aligned}$$

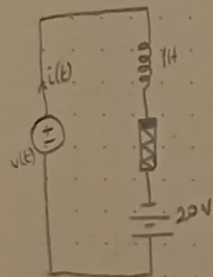
$$\frac{k_e}{R_a} = \frac{T_{stall}}{e_g} = 5$$

$$k_b = \frac{e_g}{\omega_{no-load}} = 2$$

$$\frac{G_m(s)}{E_g(s)} = \frac{5/12}{s(s + \frac{1}{12}(10 + 5 \cdot 2))} = \frac{0.417}{s(s + 1.667)}$$

$$\frac{1}{10} = \frac{G_m}{G_L} \quad \frac{G_L(s)}{E_g(s)} = \frac{0.0417}{s(s + 1.667)}$$

2.28



$i_A = 2 e^{0.14t}$ ,  $v(t)$  is small-signal source

$$v_r = 10 \ln \frac{1}{2} i_A$$

$$L \frac{di_A}{dt} + 10 \ln \frac{1}{2} i_A - 20 = v(t)$$

$i_A = 1(t)$ , in steady state  $v(t)$  is set to zero and  $v_L$  is zero. Hence  $V_r = 20V$

$$i_A = 14.78 A$$

This current is the equilibrium value of the network current. Let's call it  $i_o = 14.78 A$

$$i = i_o + \delta i \rightarrow L \frac{d(i_o + \delta i)}{dt} + 10 \ln \frac{1}{2} (i_o + \delta i) - 20 = v(t)$$

linearize

$$\ln \frac{1}{2} (i_o + \delta i) - \ln \frac{1}{2} (i_o) = \frac{d(\ln \frac{1}{2} i)}{di} \bigg|_{i=i_o} \delta i = \frac{1}{i_o} \delta i$$

$$\ln \frac{1}{2} (i_o + \delta i) = \ln \frac{1}{2} i_o + \frac{1}{i_o} \delta i$$

$$L \frac{d(i_o + \delta i)}{dt} + 10 \left( \ln \frac{1}{2} i_o + \frac{1}{i_o} \delta i \right) - 20 = v(t)$$

$$\frac{d\delta i}{dt} + 0.677 \delta i = v(t)$$

$$\delta i(s) = \frac{V(s)}{s + 0.677}$$

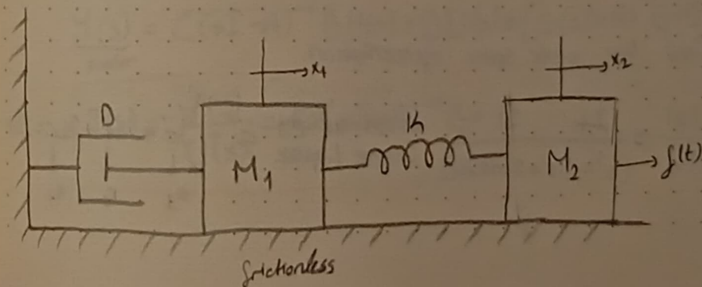
$$s \delta i(s) + 0.677 \delta i(s) = V(s)$$

$$v_L(t) = L \frac{d(i_o + \delta i)}{dt} = L \frac{d\delta i}{dt} \rightarrow V_L(s) = L s \delta i(s)$$

$$H(s) = \frac{V_L(s)}{V(s)} = \frac{s}{s + 0.677}$$

3.31

Find the state equations



$$M_1 \frac{d^2 x_1}{dt^2} + D \frac{dx_1}{dt} + kx_1 - kx_2 = 0$$

$$-kx_1 + M_2 \frac{d^2 x_2}{dt^2} + kx_2 = f(t)$$

$$\left. \begin{aligned} \ddot{x}_1 &= \dot{v}_1 \\ \ddot{x}_2 &= \dot{v}_2 \end{aligned} \right\} \text{let's assume } x_1, x_2, v_1, v_2 \text{ are state variables}$$

$$\dot{x}_1 = v_1 \quad \dot{x}_2 = v_2$$

$$\dot{v}_1 = -\frac{k}{M_1} x_1 + \frac{k}{M_1} x_2 - \frac{D}{M_1} v_1$$

$$\dot{v}_2 = \frac{k}{M_2} x_1 - \frac{k}{M_2} x_2 + \frac{f(t)}{M_2}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{v}_1 \\ \dot{v}_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{k}{M_1} & \frac{k}{M_1} & -\frac{D}{M_1} & 0 \\ \frac{k}{M_2} & -\frac{k}{M_2} & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{M_2} \end{bmatrix} f(t)$$

3.4.1

Find the state space representation.

$$\frac{C(s)}{R(s)} = \frac{24}{(s^3 + 9s^2 + 26s + 24)}$$

cross-multiply  
Inverse Laplace

$$\ddot{c} + 9\ddot{c} + 26\dot{c} + 24c = 24r$$

$\downarrow \quad \downarrow \quad \downarrow$   
 $x_3 \quad x_2 \quad x_1$

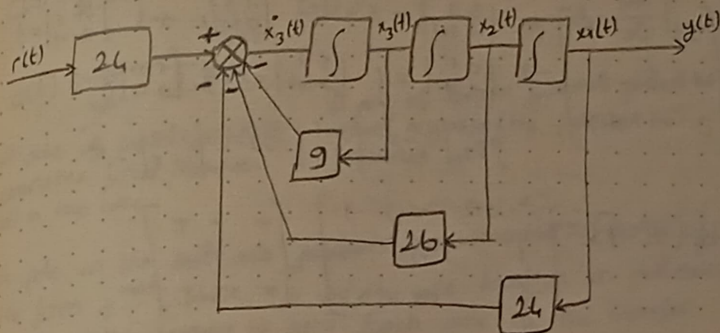
$$\dot{x}_1 = x_2 \quad \dot{x}_2 = x_3$$

$$\dot{x}_3 = 24r - 24x_1 - 26x_2 - 9x_3$$

$$y = c = x_1$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r$$

$$y = [1 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



State Space to Transfer Function

$$\begin{aligned} \dot{x} &= Ax + Bu \rightarrow sX(s) = AX(s) + BU(s) \\ y &= Cx + Du \rightarrow Y(s) = CX(s) + DU(s) \end{aligned}$$

$$X(s) = (sI - A)^{-1} BU(s)$$

$$\underbrace{Y(s)}_{\text{scalar}} = \underbrace{C(sI - A)^{-1} B}_{\text{scalar}} U(s) + \underbrace{D}_{\text{scalar}} U(s) = [C(sI - A)^{-1} B + D] U(s)$$

$$T(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1} B + D$$



3.61

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} x + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u \quad y = [1 \ 0 \ 0] x$$

Find the transfer function  $T(s) = \frac{Y(s)}{U(s)}$

$$sI - A = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 1 & 2 & s+3 \end{bmatrix}$$

$$\begin{bmatrix} s & 0 & 1 \\ -1 & s & 2 \\ 0 & -1 & s+3 \end{bmatrix} \rightarrow \begin{bmatrix} s & 2 \\ -1 & s+3 \end{bmatrix} = s^2 + 3s + 2 \quad \begin{bmatrix} -1 & 2 \\ 0 & s+3 \end{bmatrix} = -s-3 \quad \begin{bmatrix} -1 & s \\ 0 & -1 \end{bmatrix} = 1$$

$$\begin{bmatrix} s & 0 & 1 \\ -1 & s & 2 \\ 0 & -1 & s+3 \end{bmatrix} \rightarrow \begin{bmatrix} s & 1 \\ -1 & s+3 \end{bmatrix} = s^2 + 3s \quad \begin{bmatrix} s & 0 \\ 0 & -1 \end{bmatrix} = -s$$

$$\begin{bmatrix} s & 1 \\ -1 & 2 \end{bmatrix} = 2s+1 \quad \begin{bmatrix} s & 0 \\ -1 & s \end{bmatrix} = s^2$$

$$\text{Adj}(sI - A) = \begin{bmatrix} s^2 + 3s + 2 & -s-3 & 1 \\ 1 & s^2 + 3s & -s \\ -s & 2s+1 & s^2 \end{bmatrix} \times \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

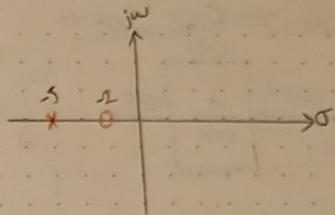
$$(sI - A)^{-1} = \frac{\text{Adj}(sI - A)}{\det(sI - A)} \quad B = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} \quad C = [1 \ 0 \ 0] \quad D = 0$$

$$C \left( \frac{\begin{bmatrix} s^2 + 3s + 2 & s+3 & 1 \\ -1 & s^2 + 3s & -s \\ -s & 2s+1 & s^2 \end{bmatrix}}{s^3 + 3s^2 + 2s + 1} \right) B + D$$

$$T(s) = \frac{10(s^2 + 3s + 2)}{s^3 + 3s^2 + 2s + 1}$$

Poles and Zeros

$$\text{Alg } \frac{1}{s} \rightarrow \frac{s+2}{s+5}$$



Let's find the unit step response of the system:  $R(s) = \mathcal{L}\{u(t)\}$

$$C(s) = \frac{s+2}{s(s+5)} = \frac{A}{s} + \frac{B}{s+5} = \frac{2/5}{s} + \frac{3/5}{s+5} \quad A = \frac{s+2}{s+5} \Big|_{s=0} = \frac{2}{5}$$

$$B = \frac{s+2}{s} \Big|_{s=-5} = \frac{3}{5}$$

$$c(t) = \frac{2}{5} + \frac{3}{5} e^{-5t}$$

A pole of input function generates forced response. (all in this case)

A pole of transfer function generates natural response (the pole at -5) generated  $e^{-5t}$ .

A pole on the real axis generates an exponential response of the form  $e^{-\alpha t}$ , where  $-\alpha$  is the pole location on real axis. It'll faster decaying to zero. Each pole of the transfer function that is on the real axis generates an exponential response that is a component of the natural response. The input pole generates the forced response.

Example

$$R(s) = 1/s \rightarrow \frac{s+3}{(s+2)(s+4)(s+5)} \quad C(s) = \frac{k_1}{s} + \frac{k_2}{s+2} + \frac{k_3}{s+4} + \frac{k_4}{s+5}$$

Forced                      Natural

$$c(t) = k_1 + k_2 e^{-2t} + k_3 e^{-4t} + k_4 e^{-5t}$$

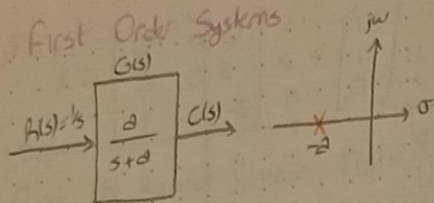
Example

A system has a transfer function  $C(s) = \frac{10(s+4)(s+6)}{(s+1)(s+7)(s+8)(s+10)}$

$R(s) = 1/s$  write  $c(t)$  in general terms.

$$c(t) = A + B e^{-t} + C e^{-7t} + D e^{-8t} + E e^{-10t}$$

## First Order Systems



$$C(s) = R(s)G(s) = \frac{a}{s(s+a)}$$

$$c(t) = 1 - e^{-at}$$

input pole at the origin generated the forced response  $c_f(t) = 1$ , and system pole at  $-a \rightarrow c_n(t) = -e^{-at}$

$a$  is the only parameter needed to describe the transient response

$$e^{-at} \Big|_{t=\frac{1}{a}} = e^{-1} = 0.37 \quad c(t) \Big|_{t=\frac{1}{a}} = 1 - e^{-at} \Big|_{t=\frac{1}{a}} = 1 - 0.37 = 0.63$$

$1/a$  is the time constant ( $\tau$ ) of the response; the time for  $e^{-at}$  to decay to 37% of its initial value.

We can call the parameter  $a$ , the exponential frequency [ $1/\tau$ ]. Since the derivative of  $e^{-at}$  is  $-a$  when  $t=0$ ,  $a$  is the initial rate of change of the exponential at  $t=0$ . Pole is located at the reciprocal of the time constant.

Rise time,  $T_r$ , is defined as the time for the waveform to go from 0.1 to 0.9 of its final value.

$$c(t) = 0.9 \quad T_r = \frac{2.3}{a} = \frac{0.1}{a} = \frac{2.2}{a}$$

$$c(t) = 0.1$$

Settling Time,  $T_s$ , is defined as the time for the response to reach and stay within 2% of its final value.

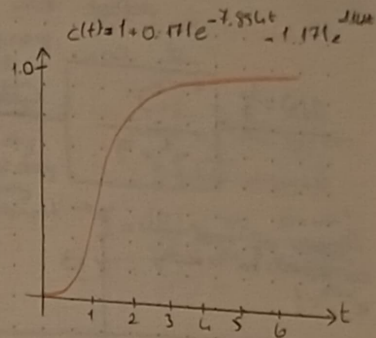
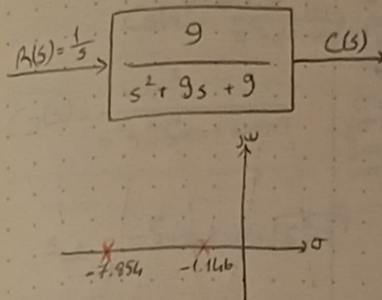
$$c(t) = 0.98 \rightarrow T_s \approx \frac{4}{a}$$

$$R(s) = \frac{1}{s} \quad C(s) = \frac{K}{s(s+a)} = \frac{K/a}{s} - \frac{K/a}{s+a}$$

$$C(s) = \frac{K}{s+a} \quad K_{1/a} = \text{loop testing value}$$

## Second Order Systems $\rightarrow$

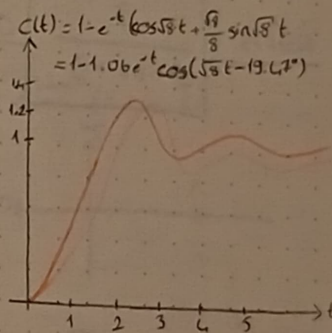
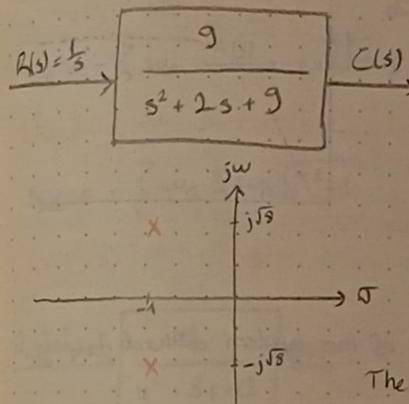
## Overdamped Response



$$C(s) = \frac{9}{s(s+7.854)(s+1.146)}$$

For this response function has two system poles on real axis that generates an exponential natural response whose frequency is equal to the pole location.

## Underdamped Response



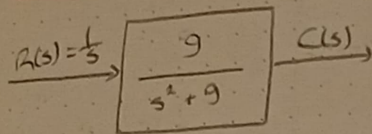
The poles that generate natural response are at  $s = -1 \pm j\sqrt{8}$   
 sinusoid's amplitude  $\rightarrow$  frequency of the oscillation

$$C(s) = \frac{9}{s(s^2 + 2s + 9)}$$

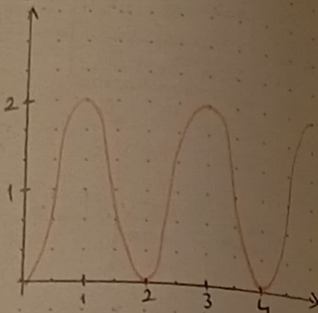
Approaches a steady-state value via a transient response that is a damped oscillation



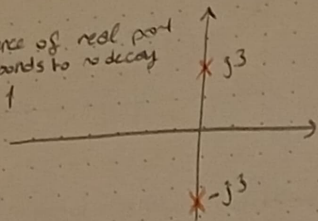
## Undamped Response



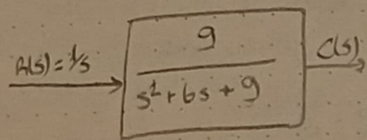
$$c(t) = 1 - \cos 3t$$



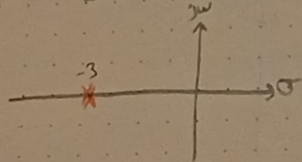
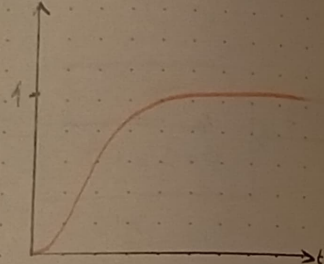
Absence of real part corresponds to no decay  
 $e^{-\sigma t} = 1$



## Critically Damped Response



$$c(t) = 1 - 3te^{-3t} - e^{-3t}$$



## Natural frequency, $\omega_n$

The frequency of oscillation of the system without damping

## Damping ratio, $\zeta$

A viable definition is comparing the exponential decay frequency (of the envelope, underdamped) to the natural frequency

$$\zeta = \frac{\text{Exponential decay frequency}}{\text{Natural frequency (rad/sec)}}$$

Consider the general second order system

$$G(s) = \frac{b}{s^2 + as + b}$$

Imaginary poles without damping  $\rightarrow a=0$   $\frac{b}{s^2 + b}$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$\sigma = -\omega_n \zeta$   
exp decay frequency underdamped

poles of the system  $\pm j\sqrt{b}$ ,  $\omega_n = \sqrt{b}$

$$\zeta = \frac{\sigma}{\omega_n} = \frac{a/2}{\omega_n} \quad b = \omega_n^2 \quad a = 2\zeta\omega_n$$

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

## Undamped ( $\zeta = 0$ )

$$s_{1,2} = \pm j\omega_n \xrightarrow{R(s)} c(t) = 1 - \cos(\omega_n t)$$

## Underdamped ( $0 < \zeta < 1$ )

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2} = -\sigma \pm j\omega_d$$

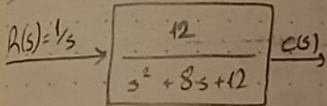
## Critically damped ( $\zeta = 1$ )

$$s_{1,2} = -\zeta\omega_n \xrightarrow{R(s)} c(t) = 1 - e^{-\omega_n t} (1 + \omega_n t)$$

## Overdamped ( $\zeta > 1$ )

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

## Example



$$a = 2\zeta\omega_n \quad \omega_n = \sqrt{b} = \sqrt{12}$$

$$a = 2\zeta 2\sqrt{3} \quad \zeta = \frac{8}{4\sqrt{3}} = \frac{2\sqrt{3}}{3}$$

$$\zeta = 1.15 > 0 \text{ overdamped}$$

$$\zeta = \frac{a}{2\sqrt{b}}$$

## Underdamped Second Order Systems

Our first objective is to define transient specifications associated with underdamped responses. Next, we relate these specifications to the pole location. Finally, tie the pole location to system parameters.

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{k_1}{s} + \frac{k_2 s + k_3}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

rest is calculation

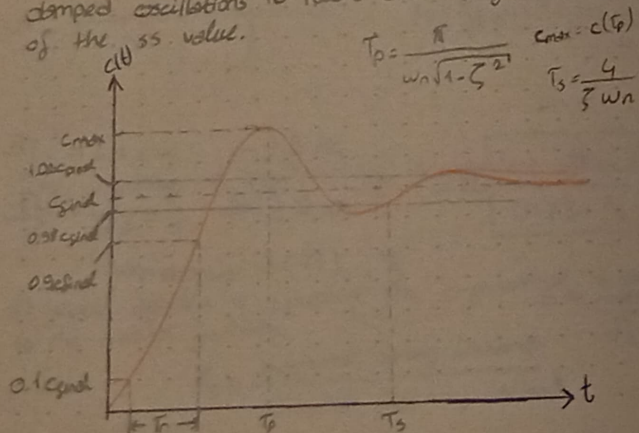
The lower the value of  $\zeta$ , the more oscillatory the response.  $\omega_n$  is a time-axis scale factor and does not affect the nature of the response other than to scale it in time.

**Rise Time  $T_r$ :** The time required for the waveform to go from 0.1 of the final value to 0.9 of the final value.

**Peak Time  $T_p$ :** The time required to reach the first (max) peak.

**Percent overshoot %OS:** The amount that the waveform overshoots the steady-state, expressed as a percentage of ss value.

**Settling time  $T_s$ :** The time required for the transient's damped oscillations to reach and stay within  $\pm 1\%$  of the ss value.



$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} \quad c_{max} = c(T_p)$$

$$T_s = \frac{4}{\zeta \omega_n}$$

Example Find  $T_p$ , %OS,  $T_s$ ,  $T_r$   $G(s) = \frac{100}{s^2 + 15s + 100}$

$$\omega_n = 10$$

$$\zeta = \frac{a}{2\omega_n} = \frac{15}{20} = 0.75$$

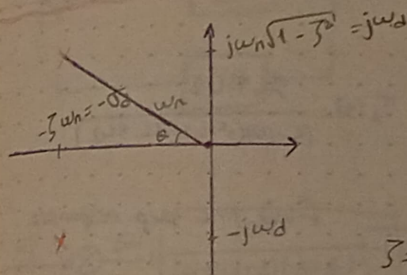
$$T_p = \frac{3.14}{10 \sqrt{1-0.75^2}} = 0.475$$

$$\%OS = \frac{c(T_p) - c_{ss}}{c_{ss}} \times 100$$

for the unit step used for our  $c(t) \rightarrow c_{ss} = 1$

$$\%OS = e^{-\left(\zeta \pi \sqrt{1-\zeta^2}\right)} \times 100 = 2.938$$

$$T_s = \frac{4}{10 \cdot 0.75} = 0.533 \text{ sec}$$

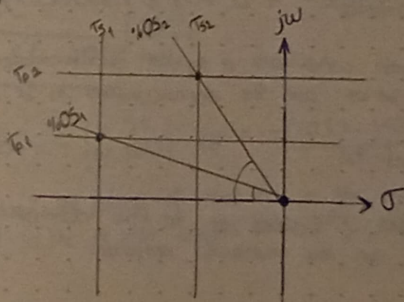


$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = \frac{\pi}{\omega_d}$$

$$T_s = \frac{4}{\zeta \omega_n} = \frac{\pi}{\sigma_d}$$

$$\zeta = \cos \theta$$

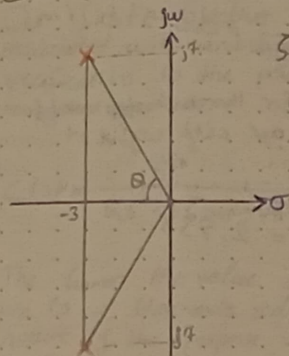
Since horizontal lines on s-plane are lines of constant imaginary value they are also lines of constant  $T_p$ . Similarly, vertical lines are constant  $T_s$  and radial lines are constant %OS.





## Example

Given the pole plot, find  $\zeta$ ,  $\omega_n$ ,  $T_p$ , %OS and  $T_s$



$$\zeta = \cos \theta = \cos(\arctan(4/3)) = 0.396$$

$$\omega_n = \sqrt{7^2 + 3^2} = 7.616$$

$$T_p = \frac{\pi}{\omega_d} = \frac{\pi}{7} = 0.449 \text{ [sec]}$$

$$\%OS = e^{-(\zeta \pi / \sqrt{1-\zeta^2})} \times 100 = 26\%$$

$$T_s = \frac{4}{\sigma_d} = \frac{4}{3} = 1.333 \text{ [sec]}$$

## Three-Pole Systems

$$T_1(s) = \frac{24.542}{s^2 + 4s + 24.542}$$

$$T_2(s) = \frac{245.42}{(s+10)(s^2 + 4s + 24.542)}$$

$$T_3(s) = \frac{73.626}{(s+3)(s^2 + 4s + 24.542)}$$

Find the step responses.

$$c_1(t) = 1 - 1.09e^{-2t} \cos(4.532t - 25.8^\circ)$$

$$c_2(t) = 1 - 0.29e^{-10t} - 1.189e^{-2t} \cos(4.532t - 53.34^\circ)$$

$$c_3(t) = 1 - 1.14e^{-3t} + 0.707e^{-2t} \cos(4.532t + 78.63^\circ)$$

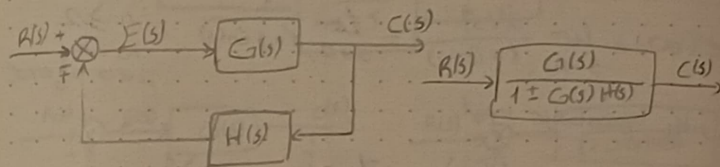
Under certain conditions, a system with more than two poles or with zeros can be approximated as a second-order system.

$$\sigma_r > |\zeta \omega_n|$$

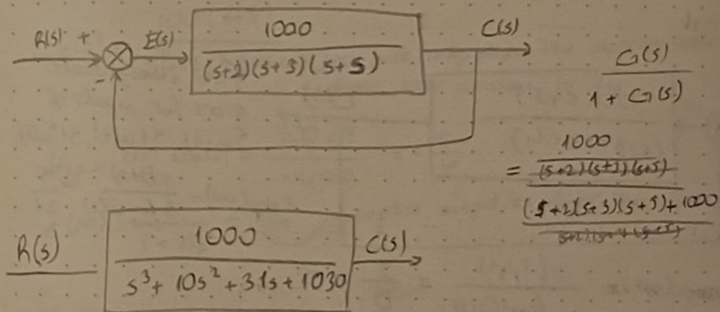
The closer the zero is to the dominant poles, the greater its effect on the transient response.

## Stability

A linear time-invariant system is stable if the natural response approaches zero as time approaches infinity. Unstable if the natural response grows without bound. Marginally stable if natural response remains constant or oscillates. A system is stable if every input yields a bounded output. (BIBO) A system is unstable if any bounded input yields an unbounded output. Stable systems have closed-loop transfer functions with poles only in the left half-plane.



## Creating a Routh Table



$$\begin{array}{c|ccc} s^3 & 1 & 31 & \\ s^2 & 10 & 1030 & \\ s^1 & -72 & & \\ s^0 & 1030 & & \end{array}$$

$$b_1 = - \left| \begin{array}{cc} 1 & 31 \\ 10 & 1030 \end{array} \right| / 10 = - \frac{(1030 - 310)}{10} = -72$$

# roots of the polynomial that are in the right-half-plane is equal to the # sign changes in the first column.

Unstable



$$e(\infty) = \lim_{s \rightarrow 0} \frac{sR(s)}{1+G(s)} \quad \left\{ \text{with step input} \rightarrow \lim_{s \rightarrow 0} \frac{s(1/s)}{1+G(s)} \right.$$

In order to have zero steady-state error we should have

$$\lim_{s \rightarrow 0} G(s) = \infty, \quad n \geq 1$$

open loop Gf

$$G(s) = \frac{(s+z_1)(s+z_2)\dots}{s^n(s+p_1)(s+p_2)\dots} \rightarrow \frac{1}{1 + \lim_{s \rightarrow 0} \frac{(s+z_1)(s+z_2)}{s^n(s+p_1)(s+p_2)}}$$

with ramp input

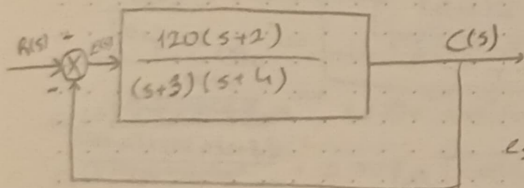
$$e(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s)}{1+G(s)} = \lim_{s \rightarrow 0} \frac{1}{s+G(s)}, \quad n \geq 2$$

to have zero ss error

with parabolic input

$$e(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^2)}{1+G(s)} = \lim_{s \rightarrow 0} \frac{1}{s^2 G(s)}, \quad n \geq 3$$

Example



Find steady-state errors for inputs of  $St(t)$ ,  $St(ult)$ ,  $St^2(ult)$ .

$$e_{step}(\infty) = \frac{s(1/s)}{1 + \lim_{s \rightarrow 0} G(s)} = \frac{5}{24}$$

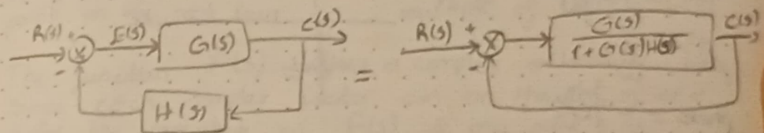
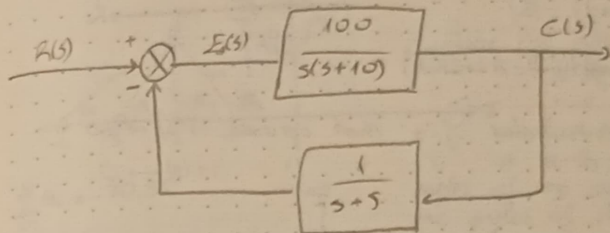
$$e_{ramp}(\infty) = \frac{s(1/s^2)}{\lim_{s \rightarrow 0} sG(s)} = \frac{5}{0} = \infty$$

$$e_{parabolic}(\infty) = \frac{s(1/s^3)}{\lim_{s \rightarrow 0} s^2 G(s)} = \frac{10}{0} = \infty$$

$$K_p = \lim_{s \rightarrow 0} G(s) \quad K_v = \lim_{s \rightarrow 0} sG(s) \quad K_a = \lim_{s \rightarrow 0} s^2 G(s)$$

$$e_{step}(\infty) = \frac{1}{1+K_p} \quad e_{ramp}(\infty) = \frac{1}{K_v} \quad e_{parabolic}(\infty) = \frac{1}{K_a}$$

**Problem:** Find the system type, the appropriate error constant associated with the system type, and the steady-state error for a unit step input.



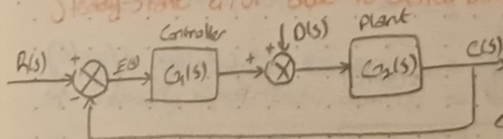
$$G_e(s) = \frac{100(s+5)}{s^3 + 15s^2 - 50s - 400}$$

no pure integration, the appropriate static error constant is  $K_p$

$$K_p = \lim_{s \rightarrow 0} G_e(s) = \frac{100 \times 5}{-400} = -\frac{5}{4}$$

$$e(\infty) = \frac{1}{1+K_p} = \frac{4}{3} \rightarrow \text{implies output is larger than input step}$$

Steady-state error due to disturbance



$$e(\infty) = -\frac{1}{\lim_{s \rightarrow 0} \frac{1}{G_2(s)} + \lim_{s \rightarrow 0} G_1(s)}$$

# BODE PLOTS

## Constant

$$H(s) = k$$

$$gain = |H(j\omega)| = |k|$$

$$phase = \arg(H(j\omega)) = \angle \tan 2(0, k)$$

a pole is a value of 's' that causes  $H(s) \rightarrow \infty$   
 a zero is " " " " " "  $H(s) \rightarrow 0$

$$H(s) = \frac{1}{s}$$

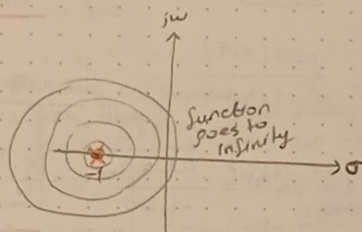
$$H(j\omega) = \frac{1}{j\omega} = -\frac{1}{\omega}j$$

$$gain = \frac{1}{\omega}$$

$$phase = -90^\circ$$

$$H(s) = \frac{1}{s+1}$$

write  $H(s)$  in form of  
 $\frac{1}{1+s/\omega_p}$  or  $1 + \frac{s}{\omega_p}$

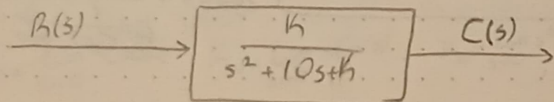


## I'm understanding Root Locus

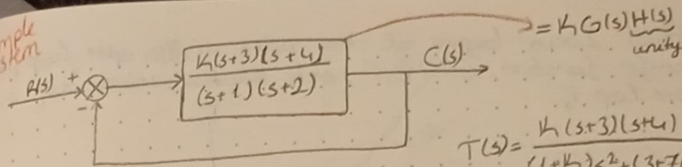
$$T(s) = \frac{kG(s)}{1+kG(s)H(s)}$$

poles  $\rightarrow k = \frac{1}{|G(s)H(s)|}$

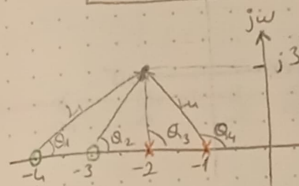
value of this must be unity  $= \frac{k}{s(s+10)} \rightarrow 9.47$  for  $s$  (yields)  $5$  for  $k$  (1)



Example system



$$T(s) = \frac{k(s+3)(s+4)}{(1+k)s^2 + (3+7k)s + (2+12k)}$$



$$\sum_{i=1}^n \theta_i = -70.55^\circ$$

Therefore,  $-2+j3$  is not a point on the root locus

Consider the point  $-2+j3$   
 if this point is a closed loop pole for a  $k$ , then the angles of the zeros minus the angles of the poles must equal an odd multiple of  $180^\circ$

angles of the zeros minus the angles of the poles must equal an odd multiple of  $180^\circ$

The function  $F(s)$  defines a complex arithmetic. Since each complex factor can be thought of as a vector, the magnitude of  $F(s)$  at any point  $s$  is

$$M = \frac{\prod \text{zero lengths}}{\prod \text{pole lengths}}$$

The angle of  $F(s)$  at any point is

$$\angle = \sum \text{zero angles} - \sum \text{pole angles}$$

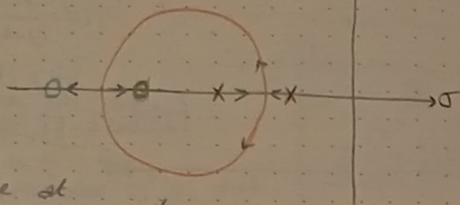
$$k = \frac{1}{|G(s)H(s)|} = \frac{1}{M} = \frac{L_3 L_4}{L_1 L_2}$$

It appears that the root locus can be obtained by sweeping through every point

in the  $s$ -plane to locate those points for which the angles, as previously described, add up to an odd multiple of  $180^\circ$

The number of branches of the root locus equals the # closed-loop poles

Root Locus begins at the poles of the open-loop transfer function,  $G(s)H(s)$ ; ends at the zeros of open-loop TF.



$$G(s) = s$$

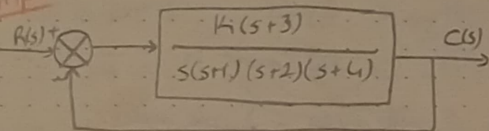
has a pole at infinity since it approaches infinity as  $s \rightarrow \infty$ .

$G(s) = 1/s$  has zero at infinity.

Consider  $K G(s)H(s) = \frac{K}{s(s+1)(s+2)} \approx \frac{K}{s \cdot s \cdot s}$

This system has three finite poles and three infinite zeros. Thus, the root locus begins at the poles and ends at infinite zeros.

Example

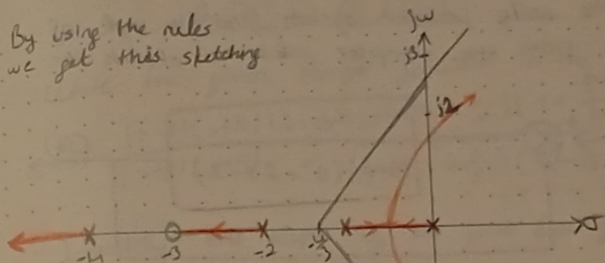


the eq. for asymptotes  $\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$

the eq. for angle  $\theta_a = \frac{(2h+1)\pi}{\# \text{finite poles} - \# \text{finite zeros}}$

$$\sigma_a = \frac{(-1-2-4) - (-3)}{4-1} = -\frac{4}{3} \quad \theta_a = \frac{(2h+1)\pi}{3}$$

By using the rules we get this sketching



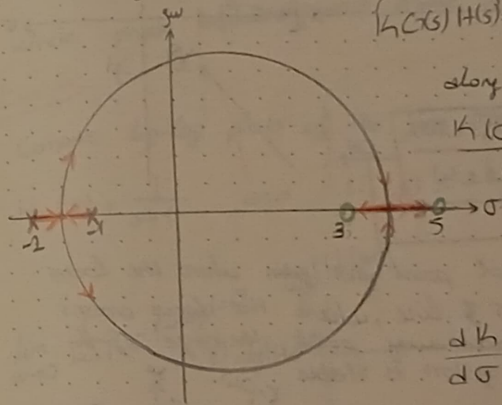
Example

Find the breakaway and break-in points for the root locus using diff calculus.

$$K G(s)H(s) = \frac{K(s-3)(s-5)}{(s+2)(s+1)}$$

along the real axis

$$\frac{K(s^2 - 8s + 15)}{s^2 + 3s + 2} = -1$$



$$K = \frac{-(s^2 + 3s + 2)}{s^2 - 8s + 15}$$

$\frac{dK}{ds} = 0$  gives us the  
 $\sigma = -1.65$  breakaway  
 $\sigma = 3.82$  break-in

$$\sum \frac{1}{\sigma + p_i} = \sum \frac{1}{\sigma + z_i}$$

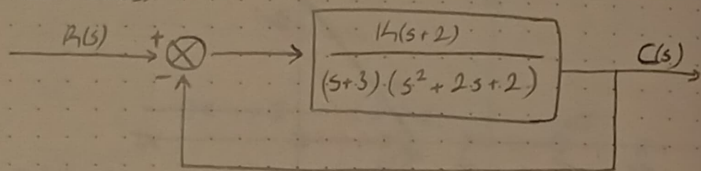
$$\frac{1}{\sigma-3} + \frac{1}{\sigma-5} = \frac{1}{\sigma+2} + \frac{1}{\sigma+1}$$

$$11\sigma^2 + 26\sigma - 61 = 0 \text{ yields the same}$$



Example

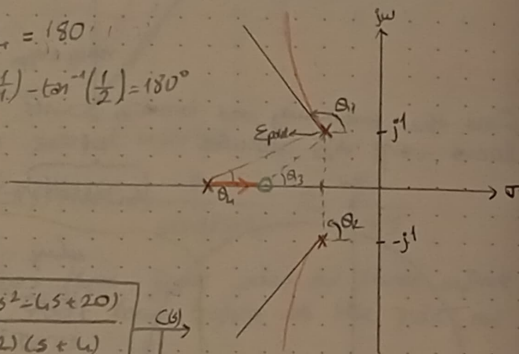
Given the unity feedback system, find the angle of departure from the complex poles and sketch the root locus.



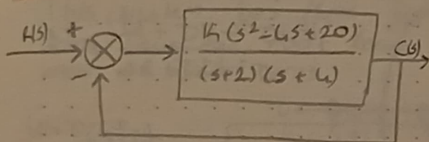
$$\angle_1 - \angle_2 + \angle_3 - \angle_4 = 180^\circ$$

$$\angle_1 - 90 + \tan^{-1}\left(\frac{1}{1}\right) - \tan^{-1}\left(\frac{1}{2}\right) = 180^\circ$$

$$\angle_1 = 109.4^\circ$$



Example



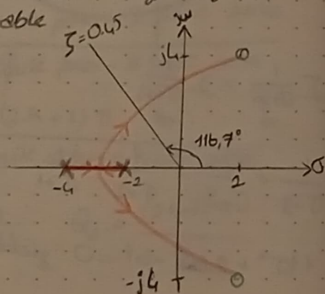
Sketch the exact point and gain where the locus crosses the 0.45  $\zeta$  line, where the locus crosses the  $j\omega$ -axis, the breakaway point, the range of  $K$  within which the system is stable.

$$s^2 - 4s + 20 = 0$$

$$s = 2 \pm j4 \text{ zeros}$$

$$s = -2, -4 \text{ poles}$$

$$\theta = 180 - \cos^{-1} 0.45 = 116.7^\circ$$



## BODE PLOTS

1- Write the given transfer function in the standard form

$$G(s) = \frac{(s+a)(s+b)}{(s+p)(s+q)} = \frac{ab(1+\frac{s}{a})(1+\frac{s}{b})}{pq(1+\frac{s}{p})(1+\frac{s}{q})} \quad \frac{ab}{pq} = K$$

2- Identify the slope of the first line for the bode based on poles and zeros.

Find the gain of first line at  $\omega = 1 \text{ rad/sec}$

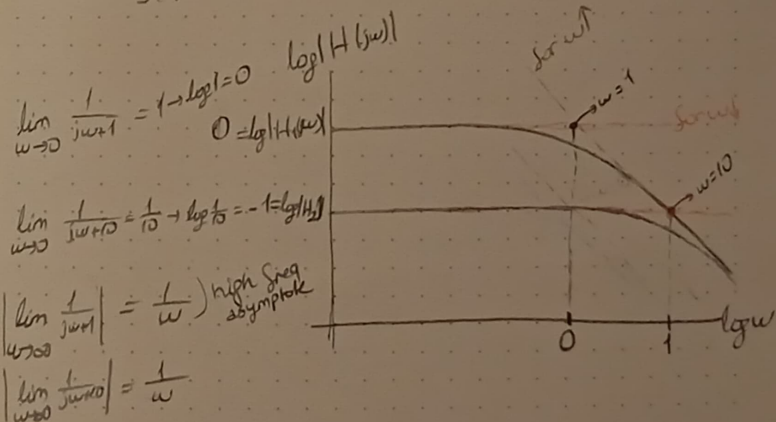
Write all the corner frequencies in ascending order and define the slope.

Write phase equation and make a table.

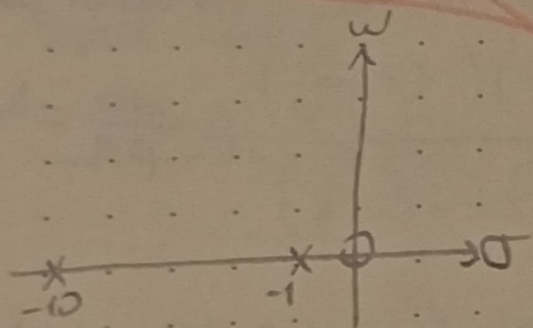
Example

Compare log-log plots of the freq responses

$$H_1(s) = \frac{1}{s+1} \quad \text{and} \quad H_2(s) = \frac{1}{s+10}$$



$$H(s) = \frac{s}{(s+1)(s+10)}$$



$$\log \left| \frac{j\omega}{(j\omega+1)(j\omega+10)} \right|$$

